5

Signals, Systems, and Transforms

This chapter focuses on **transforms**: a family of techniques used to study a series of values, or **signals**. Transforms give insight into the nature of signals and help design effective **systems** to manipulate signals. Transforms form the basis for the study of control theory, discussed in more detail in Chapter 8.

We begin by covering background material on complex numbers and Euler's formula. We then discuss types of signals and systems. Subsequently, the bulk of the chapter is devoted to the study of Fourier series, Fourier transforms, Laplace transforms, and the Z transform.

5.1 Background

5.1.1 Sinusoids

A communication network is formed from communication links. A widely used approach to transferring information over a wired or wireless link is to modify an underlying continuous **carrier** signal to represent the symbol values 0 and 1. In nearly all cases of interest, the carrier signal is **sinusoidal**. That is, it is associated with a time period T such that at time t, the amplitude of the signal is described by the function $\cos\left(\frac{2\pi t}{T}\right)$, as shown in Figure 5.1.

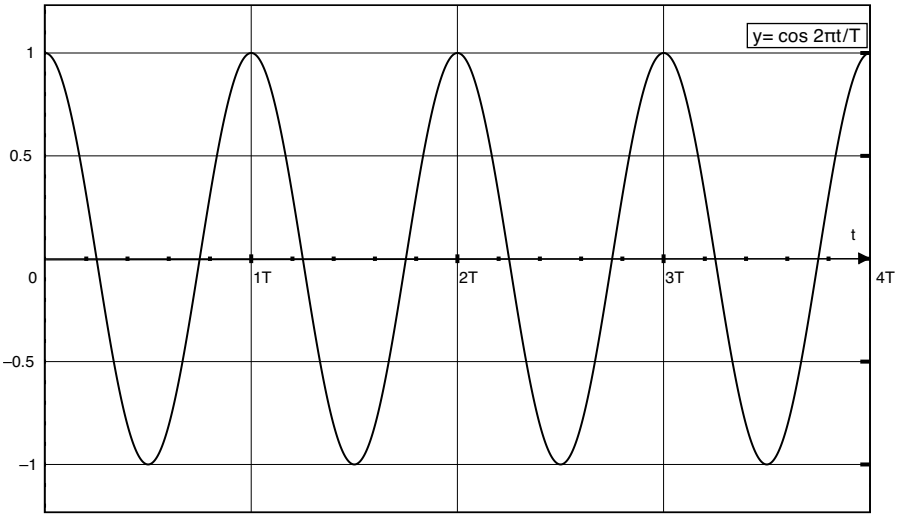


Figure 5.1 A typical carrier wave described by a sinusoidal curve

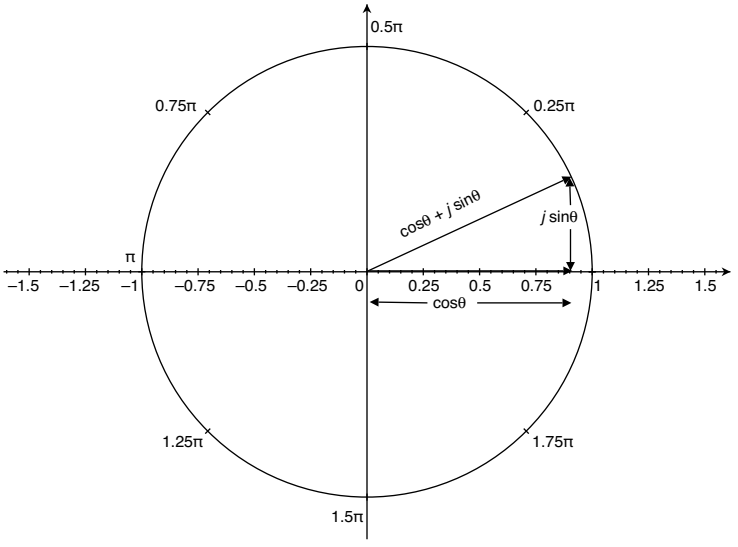
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How can we generate such a time-varying signal? Consider a disk with a radius of one unit that has an axle at its center. Suppose that it rotates in the counterclockwise direction with a uniform angular velocity and with a period of T seconds. Let us paint a small dot on its rim and draw a vector from the center to this dot. The angle that this rotating vector makes with the X-axis at time t is denoted by θ and given by $\theta = \frac{2\pi t}{T}$ radians (Figure 5.2). At time t, the distance of the dot from the Y-axis is given by $\cos(\theta) = \cos\left(\frac{2\pi t}{T}\right)$, which is precisely the equation describing a sinusoidal carrier. In other words, the unidimensional *time-varying* signal strength of a sinusoidal carrier can be thought of as a projection of a *fixed-length* two-dimensional rotating vector on the X-axis. This warrants a more careful study of the mathematics of rotating two-dimensional vectors.

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5.1.2 Complex Numbers

Recall that any vector on a two-dimensional plane can be written as the vector sum of two *basis* vectors, one lying on the X-axis and one parallel to the Y-axis. Let us consider how to represent these basis vectors. A real value x on the number line naturally corresponds to a vector lying on the X-axis drawn from (0,0) to (0, x). Thus, the first of these basis vectors corresponds to, and can be represented by, a



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Figure 5.2 Generating a sinusoid from a rotating vector

real number. How can we represent the second vector? Intuitively, we need a way to "raise" a vector off the real axis. Suppose that we denote the operation "rotate a vector counterclockwise by 90 degrees about the origin" by the letter j.[1] Then, the vector $j5$, for example, corresponds to a vector of magnitude 5 lying on the Y-axis. Given this operation, we can denote a vector on the unit disk making an angle θ with the X-axis as the vector sum $\cos(\theta) + j\sin(\theta)$, as shown in Figure 5.2.

We denote the application of the rotation operator j twice in a row like this: $j.j$, or j^2. Note that the application of this combined operator to the vector 1, which is the vector from (0,0) to (0,1), results in the vector –1. We represent this by the equation

$$j.j.1 = j^2.1 = -1 \qquad \textbf{(EQ 5.1)}$$

With a slight abuse of notation, we can take the square root of either side to get

$$j = \sqrt{-1} \qquad \textbf{(EQ 5.2)}$$

revealing j to be the unit imaginary vector. It is unfortunate that most students encounter j as a ghostly, ethereal entity rather than as the far more intuitive 90-degree rotation operator.

1. Mathematicians and engineers differ in the symbol used for this operation. Engineers use j rather than the mathematician's i because by universal engineering convention, the symbol i represents an electric current.

Given the basis vectors 1 and j, it is easy to see that we can represent a vector from the origin to any point (a,b) on the plane as the vector sum $a + jb$. The corresponding tuple (a,b) is called a **complex** number, with a as its **real** part and b as its **imaginary** part. If $z = a+jb$ is a complex number, we denote $a = \text{Re}(z)$; $b = \text{Im}(z)$. We denote by z^* the **complex conjugate** of z, defined as $z^* = a - jb$.

5.1.3 Euler's Formula

Given the importance of rotating vectors, also called **phasors**, it is desirable to compactly represent the current position of the vector on the unit disk. Of course, this can be easily specified by the single value θ. By tradition, and for sound mathematical reasons that we will see later, we instead represent a rotating vector of magnitude c making an angle θ with the X-axis by using the notation $ce^{j\theta}$. This notation is rather counterintuitive! What does it mean to raise e to an imaginary power? The answer is: It does not matter. All that matters is that the following formula, known as **Euler's formula**,[2] has been found to be consistent with the foundations of real analysis:



$$e^{j\theta} = \cos\theta + j\sin\theta \qquad \textbf{(EQ 5.3)}$$

EXAMPLE 5.1: EULER'S FORMULA

Use Euler's formula to represent $e^{j\frac{\pi}{2}}$ in the form $a + jb$.

Solution:

By definition, $e^{j\frac{\pi}{2}} = \cos\left(\frac{\pi}{2}\right) + j\sin\left(\frac{\pi}{2}\right) = j$.

EXAMPLE 5.2: EULER'S FORMULA

Use Euler's formula to represent $e^{-j\frac{\pi}{10}} + 2e^{-j\frac{2\pi}{10}}$ in the form $a + jb$.

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2. The Euler formula has been called the most beautiful formula in mathematics, especially when written for the special value of $\theta = \pi$, where, using standard mathematical convention, it reduces to

$$e^{i\pi} + 1 = 0$$

relating five elementary quantities in mathematics: *e, i,* 0, 1, and π.

Solution:

$$
\begin{aligned}
& e^{-j\frac{\pi}{10}} + 2e^{-j\frac{2\pi}{10}} \\
&= \left(\cos\left(-\frac{\pi}{10}\right) + j\sin\left(-\frac{\pi}{10}\right)\right) + 2\left(\cos\left(-\frac{2\pi}{10}\right) + j\sin\left(-\frac{2\pi}{10}\right)\right) \\
&= \left(\cos\left(\frac{\pi}{10}\right) + 2\cos\left(\frac{2\pi}{10}\right)\right) - j\left(\sin\left(\frac{\pi}{10}\right) + 2\sin\left(\frac{2\pi}{10}\right)\right) \\
&= (0.95 + 1.618) - j(0.309 + 1.175) \\
&= 2.568 + j1.484
\end{aligned}
$$

We have already seen that the projection of a phasor $ce^{j\theta}$ on the X-axis is given by $c\cos(\theta)$, which is the first, or real, component in the expansion of $ce^{j\theta}$ using Euler's formula. We denote this as $\mathrm{Re}(ce^{j\theta})$. Recalling that $\theta = \frac{2\pi t}{T}$, the strength of a carrier signal at time t is therefore given by

$$\mathrm{Re}(ce^{j\left(\frac{2\pi t}{T}\right)}).$$

The factor $\frac{2\pi}{T}$ occurs so frequently in the study of rotating vectors that it has its own symbol, ω, where $\omega = \frac{2\pi}{T}$. The symbol ω has a physical interpretation as the angular frequency of rotation and has units of radians/second. With the notation and concepts built up so far, we can succinctly denote a carrier signal as $\mathrm{Re}(ce^{j\omega t})$.

We have seen two ways of representing a complex number: as a sum $a + jb$, or as a phasor $ce^{j\theta}$. It is straightforward to go from one representation to the other, using the following formulae:

$$
\begin{aligned}
a &= c\cos\theta \qquad & c &= \sqrt{a^2 + b^2} \\
b &= c\sin\theta \qquad & \theta &= \mathrm{atan}\left(\frac{b}{a}\right)
\end{aligned}
\qquad \textbf{(EQ 5.4)}
$$

Note that we have arbitrarily described the phasor as rotation in the counterclockwise direction. A sinusoidal carrier can equally well be generated by a phasor rotating in the counterclockwise direction. To distinguish between the two, we denote the angular frequency of a phasor rotating in the clockwise direction as a negative frequency. Thus, a phasor with a frequency of -20π radians/second is rotating 10 times a second in the clockwise direction, and the second hand of a clock has a frequency of $-\frac{\pi}{30}$ radians/second.

EXAMPLE 5.3: KEYING

We now consider how to mathematically represent the transmission of information on a link by modification of the phasor $ce^{j\omega t}$. This is also called **keying**. For simplicity, focus on the problem of sending just one bit: either a 0 or a 1.

In **amplitude shift keying**, we choose two phasor magnitudes c_1 and c_2. To send a 0, the transmitter transmits a signal $c_1\text{Re}(e^{j\omega t})$; to send a 1, it transmits a signal $c_2\text{Re}(e^{j\omega t})$. In practice, this is accomplished by an electronic circuit that sets the degree of amplification of the carrier signal to either c_1 or c_2. If either of the constants c_1 and c_2 is 0, this is also called **on-off keying**.

In **frequency shift keying**, we choose two phasor frequencies ω_1 and ω_2. To send a 0, the transmitter transmits a signal $c\text{Re}(e^{j\omega_1 t})$; to send a 1, it transmits a signal $c\text{Re}(e^{j\omega_2 t})$. In practice, this is accomplished by either coupling the output signal to one of two oscillators tuned to the frequencies ω_1 and ω_2 or sending control signals to a voltage-controlled oscillator.

The **phase** of a sinusoid at time t is the angle it makes with the X-axis at that time. With **phase shift keying**, the transmitter and receiver are assumed to be **phase-locked**; that is, they each have local oscillators that have identical frequencies and phases. The transmitter sends a 0 or a 1 by modifying the phase of the carrier signal. With **binary phase shift keying**, for example, the transmitter transmits the unmodified carrier signal $ce^{j\omega t}$ to send a 0 and $-ce^{j\omega t}$, which is a signal that is exactly 180 degrees out of phase with the receiver's local oscillator, to send a 1. This general idea can be easily extended. For example, with **quadrature phase shift keying**, the transmitter shifts the transmitting phasor's phase by 0, 90, 180, or 270 degrees, using the phasors

$$ce^{j\omega t},\ ce^{j\left(\omega t+\frac{\pi}{2}\right)},\ ce^{j(\omega t+\pi)},\ \text{and}\ ce^{j\left(\omega t+\frac{3\pi}{2}\right)}.$$

These phases allow the receiver to extract *two* bits for each received signal. In practice, phase shift keying is accomplished by slightly delaying the transmitted carrier signal so that it is received out of phase.

One might ask: What is the limit to phase shift keying? Would it be possible to send larger and larger numbers of bits by using more fine-grained phase shifting? Could we send, for example, 32 bits at a time by using 65,536 phases? The answer is that as we increase the number of phase levels, even a slight distortion of a signal along the path from the transmitter to the receiver corrupts the signal. Therefore, the limit to the *capacity* of the path comes from its inherent *noise*. The relationship between the capacity of a channel and its noise characteristics form the basis of *information theory*, which is discussed in Chapter 9.

It is sometimes useful to express sinusoids as complex exponentials. We can do so as follows. First, recall that $\cos(-\theta) = \cos\theta$ and $\sin(-\theta) = -\sin\theta$. Now, because $e^{j\theta} = \cos\theta + j\sin\theta$, it follows that $e^{-(j\theta)} = e^{j(-\theta)} = \cos(-\theta) + j\sin(-\theta) = \cos\theta - j\sin\theta$. Therefore, $e^{j\theta} + e^{-j\theta} = 2\cos\theta$, so that

$$\cos\theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta}) \quad \textbf{(EQ 5.5)}$$

Similarly,

$$\sin\theta = \frac{1}{2j}(e^{j\theta} - e^{-j\theta}) \quad \textbf{(EQ 5.6)}$$

5.1.4 Discrete-Time Convolution and the Impulse Function

Consider the discrete-time function $x(t)$, where t is a non-negative integer, that represents the number of HTTP requests made to a Web server over intervals of 1 second. Suppose that this function, over an interval of 10 seconds, is given by $x(t)$, $0 \le t \le 9$ = 1, 3, 5, 2, 5, 8, 7, 3, 9, 4. That is, in the 0th interval, the Web server receives $x(0) = 1$ request; in the first interval, it receives $x(1) = 3$ requests; in the second interval, it receives $x(2) = 5$ requests; and so on. The histogram of this function is shown in Figure 5.3(a).

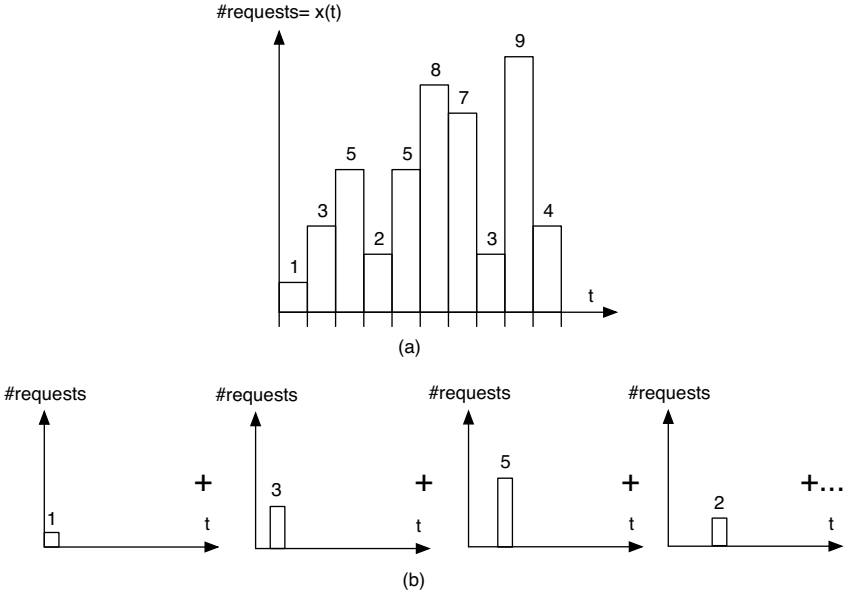


Figure 5.3 Decomposing a discrete-time function

The same histogram can also be represented as the sum of a set of smaller histograms, one per time interval, as shown in Figure 5.3(b). Each smaller histogram is the product of two functions, $x(t)$ and a selector function $sel(t, \tau)$, which is defined as $sel(t, \tau) = 1$ when $t = \tau$ and = 0 otherwise. For instance, the first small histogram corresponds to $x(t)$*$sel(t,0)$ because it is zero everywhere, except when $t = 0$, when $x(t)$ takes the value $x(0)$ and $sel(t,0)$ is 1. Similarly, the second histogram corresponds to $x(t)$*$sel(t,1)$ and so on. Because their sum is $x(t)$, we can write

$$x(t) = \sum_{\tau=0}^{\infty} x(t)sel(t, \tau) \quad \textbf{(EQ 5.7)}$$

We make a small digression to consider the relationship between the function $x(t)$ and the function $x(t - a)$. For concreteness, consider the function $x(t)$ shown in Figure 5.3(a) and let $a = 2$. The function $x(t - a)$ is undefined for times before time 2, and, for convenience, we define it to have the value 0. At time 2, $x(t - a) = x(2 - 2) = x(0) = 1$. At time 3, $x(t - a) = x(1) = 3$ and so on. We plot this in Figure 5.4.

Note that $x(t - 2)$ is just $x(t)$ shifted two steps to the right. A little thought shows that, in general, the time series $x(t - a)$ is the time series $x(t)$ shifted by a time interval a to the right.

Now, consider the function $\delta(t)$, also called the **impulse function**, defined as $\delta(0) = 1$ and $\delta(t) = 0$ for all other values of t. We see that, trivially, $sel(t, 0) = \delta(t)$. From our observation about time-shifting, we can define $sel(t, 1) = \delta(t - 1)$, which is the one-step time-shifted version of $\delta(t)$. Generalizing, we see that

$$sel(t, \tau) = \delta(t - \tau) \quad \textbf{(EQ 5.8)}$$

Therefore, we can rewrite Equation 5.7 as

$$x(t) = \sum_{\tau=0}^{\infty} x(t)\delta(t - \tau) \quad \textbf{(EQ 5.9)}$$

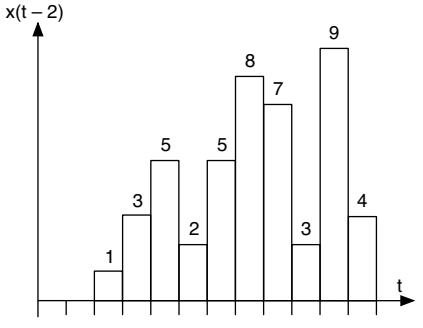


Figure 5.4 Time-shifting a time series

We have already seen that for each term in the summation, the value of the product of $x(t)$ and $\delta(t-\tau)$ is zero except when $\tau = t$, so we can also write the summation as

$$x(t) = \sum_{\tau=0}^{\infty} x(\tau)\delta(t-\tau)$$

Finally, for functions $x(t)$ that are defined over both the positive and negative values of t, we can generalize the summation over the entire number line to get

$$x(t) = \sum_{\tau=-\infty}^{\infty} x(\tau)\delta(t-\tau) \quad \textbf{(EQ 5.10)}$$

This summation is called the **convolution** of $x(t)$ with $\delta(t)$ and is denoted $x(t) \otimes \delta(t)$.[3]

In general, the convolution of two discrete functions $x(t)$ and $y(t)$ is defined as

$$x(t) \otimes y(t) = \sum_{\tau=-\infty}^{\infty} x(\tau)y(t-\tau) \quad \textbf{(EQ 5.11)}$$

Note that *each* value of the convolution of $x(t)$ and $y(t)$ (i.e., at time t) is the result of adding *all* product pairs $x(\tau)$ and $y(t-\tau)$. This is clearly a computationally expensive operation. Moreover, we need to be careful in dealing with values of $x(t)$ and $y(t)$ that lie outside their range of definition.

EXAMPLE 5.4: DISCRETE-TIME CONVOLUTION

Compute the convolution of $x(t)$, $0 \le t \le 9$ = 1, 3, 5, 2, 5, 8, 7, 3, 9, 4 and $y(t)$, $0 \le t \le 9$ = 3, 1, 7, 4, 5, 9, 7, 1, 3, 8.

Solution:

By convention, we assume that both functions are zero outside their range of definition. Let $z(t) = x(t) \otimes y(t)$: $z(0) = \sum_{\tau=-\infty}^{\infty} x(\tau)y(-\tau)$. The only value of τ for which both $x(t)$ and $y(t)$ are nonzero is $\tau = 0$, so the sum reduces to $x(0)*y(0) = 3$. $z(1) = \sum_{\tau=-\infty}^{\infty} x(\tau)y(1-\tau)$. The only values of τ for which both $x(t)$

3. In electrical engineering texts, convolution is represented by *. Unfortunately, this is universally used in computer science to denote multiplication, thus our choice.

and $y(t)$ are nonzero are $\tau = 0, 1$ so the sum reduces to $x(0) * y(1) + x(1) * y(0) = 1 * 1 + 3 * 3 = 10$. It is left as an exercise to the reader to show that $z(2) = 7 + 3 + 15 = 25$, $z(3) = 4 + 21 + 5 + 6 = 36$, $z(4) = 5 + 12 + 35 + 2 + 15 = 69$,....

5.1.5 Continuous-Time Convolution and the Dirac Delta Function

The definition of convolution can be extended to continuous-time functions. We start, as before, with the definition of a selector function with a parameter τ that picks out the value of $x(t)$ at time τ. Thus, we would like to define a function that is zero everywhere except at a single point, where it has a value of 1. Obviously, such a function is discontinuous at that point, which makes it mathematically troublesome. To avoid this problem, we define the selector function as the limiting case of a rectangular function of unit area defined over a support of length T and with height $1/T$ as $T \to 0$. At the limit, this represents an infinitely tall rectangle that nevertheless has a finite area! Nevertheless, this function is continuous everywhere except at the limit, so that its properties at the limit can be extrapolated from those of the limiting series. Setting aside a mathematically rigorous development, which is beyond the scope of this text, we define the continuous-time impulse function by

$$\int_{-\infty}^{\infty} \delta(t)dt = 1$$
$$\delta(t) = 0 \quad \text{for } t \neq 0 \qquad \textbf{(EQ 5.12)}$$

Intuitively, a continuous-time impulse function has a unit mass concentrated entirely at $t = 0$. It was first defined by the physicist P. A. M. Dirac and therefore is also called the **Dirac delta** in his honor. The Dirac delta is represented graphically by a vertical arrow at the origin. The integral of the delta function is called the **unit-step function** and written $u(t)$. Note that $u(t)$ is zero for $t < 0$ and 1 for $t > 0$. Its value at zero is undefined (Figure 5.5).

For any function $x(t)$ that is continuous in the neighborhood of the origin, the delta function acts as a selector, so that

$$x(t)\delta(t) = x(0)\delta(t) \qquad \textbf{(EQ 5.13)}$$

We interpret this to mean that the multiplication of $x(t)$ with the delta function results in a delta function whose strength is the value of $x(t)$ at the origin. (Compare this with the discrete-time selector function.) Similarly,

$$x(t)\delta(t-\tau) = x(\tau)\delta(t-\tau) \qquad \textbf{(EQ 5.14)}$$

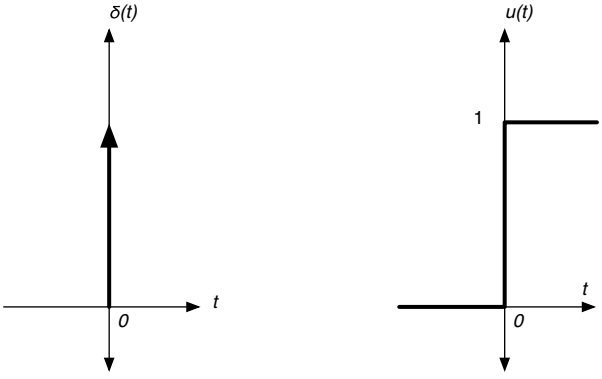


Figure 5.5 The Dirac delta and its integral, the unit-step function

Moreover, analogous to Equation 5.10, we have

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau$$
$$x(\tau) = \int_{-\infty}^{\infty} x(t)\delta(t-\tau)dt \qquad \textbf{(EQ 5.15)}$$

where the second equation can be obtained by swapping the variables t and τ.

Finally, we can define the convolution of any two continuous-time functions $x(t)$ and $y(t)$ as

$$x(t) \otimes y(t) = \int_{-\infty}^{\infty} x(\tau)y(t-\tau)d\tau \qquad \textbf{(EQ 5.16)}$$

EXAMPLE 5.5: CONTINUOUS-TIME CONVOLUTION

Compute the convolution of the functions $x(t)$ and $y(t)$ defined graphically in Figure 5.6.

Solution:

By convention, we assume that both functions are zero outside their range of definition. Thus,

$$x(\tau) \neq 0 \text{ in the range } -1 \leq \tau \leq 1$$
$$y(t-\tau) \neq 0 \text{ in the range } -1 \leq (t-\tau) \leq 1 \Rightarrow t-1 \leq \tau \leq t+1$$

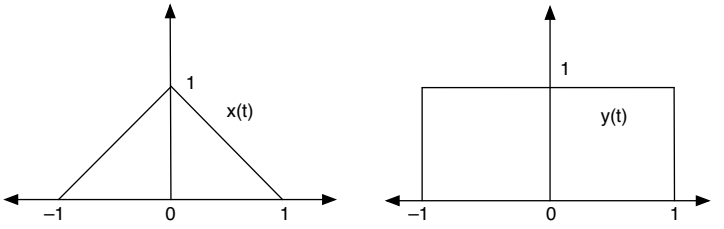


Figure 5.6 Example continuous-time functions for convolution

Therefore, both $x(t)$ and $y(t)$ are simultaneously nonzero only in the range $max(t-1,-1) \le \tau \le min(t+1,1)$, and

$$x(t) \otimes y(t) = \int_{-\infty}^{\infty} x(\tau)y(t-\tau)d\tau = \int_{max(t-1,-1)}^{min(t+1,1)} x(\tau)y(t-\tau)d\tau$$

It is necessary to evaluate this integral separately in the ranges $[-\infty,-2]$, $[-2,-1]$, $[-1,0]$, $[0,1]$, $[1,2]$, and $[2,\infty]$. In all the ranges, $y(t) = 1$, but the limits of integration are modified as follows:

$$x(t) \otimes y(t) = \begin{cases} 0 & -\infty \le t \le -2 \\ \int_{-1}^{(t+1)} (1+\tau)d\tau & -2 \le t \le -1 \\ \int_{0}^{(t+1)} (1-\tau)d\tau + \frac{1}{2} & -1 \le t \le 0 \\ \int_{(t-1)}^{0} (1+\tau)d\tau + \frac{1}{2} & 0 \le t \le 1 \\ \int_{t-1}^{1} (1-\tau)d\tau & 1 \le t \le 2 \\ 0 & 2 \le t \le \infty \end{cases}$$

$$x(t) \otimes y(t) = \begin{cases} 0 & -\infty \le t \le -2 \\ \frac{(t+2)^2}{2} & -2 \le t \le -1 \\ 1 - \frac{t^2}{2} & -1 \le t \le 0 \\ 1 - \frac{t^2}{2} & 0 \le t \le 1 \\ \frac{(t-2)^2}{2} & 1 \le t \le 2 \\ 0 & 2 \le t \le \infty \end{cases}$$

Note that, in this case, the convolution changes its form across the entire range. Therefore, it is a tricky task to compute it. As we will see later, the use of transforms greatly reduces the complexity of computing the convolution of two functions.

5.2 Signals


6:19

A signal is a set of data, usually a function of time. We denote a signal as $x(t)$, where t is nominally time. We have already seen two types of signals: **discrete** signals, where t takes integer values, and **continuous** signals, where t takes real values. In contrast, **digital** signals are those where the signal $x(t)$ takes on one of a quantized set of values, typically 0 and 1, and **analog** signals are those where the signal $x(t)$ takes on real values.

Some signals are **periodic**, so that their values repeat after every time period T. The smallest such period is called the **fundamental period** of the signal. Signals that do not have such a period are called **aperiodic**.

Most signals exist for only a limited period in time and are called **time-limited** signals. Others can be modeled as eternal signals and are called **time-unlimited** signals.

Given a signal $x(t)$, we define a **time-shifted** signal that is time-shifted a time units to the right (i.e., it is delayed by a time units as compared to the original signal) to be $x(t - a)$. Similarly, we define a **time-scaled** signal where the signal is expanded or compressed in time by the scalar value a to be $x(at)$.

Here are some examples of signals that arise in computer networks:

- The series of packet round trip times measured at a network source (discrete, analog, aperiodic, time-limited)
- The buffer occupancy over time at a router (continuous, digital, aperiodic, time-unlimited)
- The samples collected at an analog-to-digital transducer during a voice call (discrete, digital, aperiodic, time-limited)
- The physical carrier signal on a wired or wireless link (continuous, analog, periodic, time-unlimited)
- The number of HTTP requests received at a Web server over intervals of 1 second (discrete, digital, aperiodic, time-unlimited)
- The number of routing updates received by a border gateway protocol (BGP) daemon at a router over intervals of 5 minutes (discrete, digital, aperiodic, time-unlimited)

Digital systems, such as computer networks, approximate a continuous signal by a discrete signal whose time instants are sufficiently closely spaced and an analog signal by a digital signal whose quantization levels are sufficiently closely spaced. We will study what is precisely meant by "sufficiently closely spaced" later in this chapter. For the remainder of this chapter, rather than these specific signals, we will consider signals in the abstract.

5.2.1 The Complex Exponential Signal


10:08

A signal that frequently crops up in the study of transforms is the complex exponential signal denoted ke^{st}, where k is a real number and s is a complex quantity that can be written as $s = \sigma + j\omega$. We can expand ke^{st} as $ke^{st} = ke^{(\sigma+j\omega)t} = ke^{\sigma t}e^{j\omega t} = ke^{\sigma t}(\cos\omega t + j\sin\omega t) = ke^{\sigma t}\cos\omega t + jke^{\sigma t}\sin\omega t$. The real and imaginary components of this signal are both exponentially modulated sinusoids having frequency ω. With carefully chosen values of σ, ω, and k, this expression and its conjugate can represent a variety of seemingly unrelated signals:

- When $s = 0$, this reduces to the constant k (Figure 5.7(a)).
- When $\omega = 0$, this reduces to the real monotone exponential $ke^{\sigma t}$ (Figure 5.7(b)).
- When $\sigma = 0$, setting $s = \pm j\omega$ gives us the sum

  $$k((\cos\omega t + j\sin\omega t) + (\cos\omega t - j\sin\omega t)) = 2k\cos\omega t,$$

  which is a real sinusoid scaled by a factor k (Figure 5.7(c)).
- When $\sigma \neq 0$, setting $s = \sigma \pm j\omega$ gives us the sum $2ke^{\sigma t}\cos\omega t$, which is a scaled real sinusoid that is modulated by a monotone exponential (Figure 5.7(d)).
- When $\sigma = 0$, the expression reduces to $k(\cos\omega t + j\sin\omega t)$, which is a helix oriented along the t axis, whose projections on the real-t and imaginary-t planes are sinusoids (Figure 5.8(a)).
- In the general case, the function represents an exponentially modulated helix oriented along the t axis, whose projections on the real-t and imaginary-t planes are exponentially modulated sinusoids (Figure 5.8(b)).

It is worth studying these figures carefully because they provide deep insight into the nature of a complex exponential that will greatly help in understanding the nature of transforms.

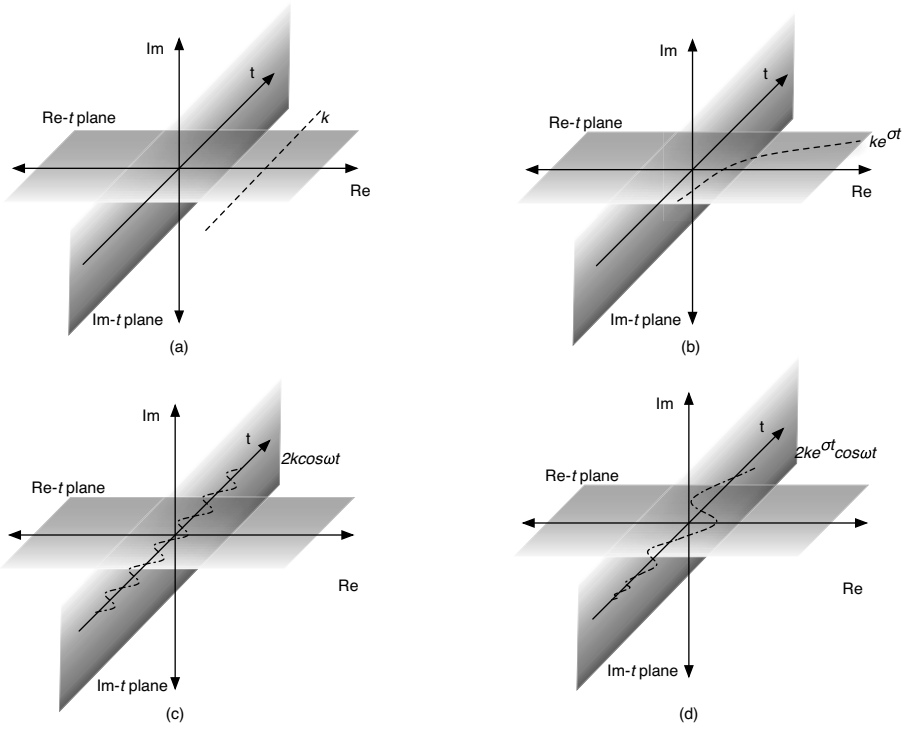


Figure 5.7 Elementary cases of the complex exponential signal

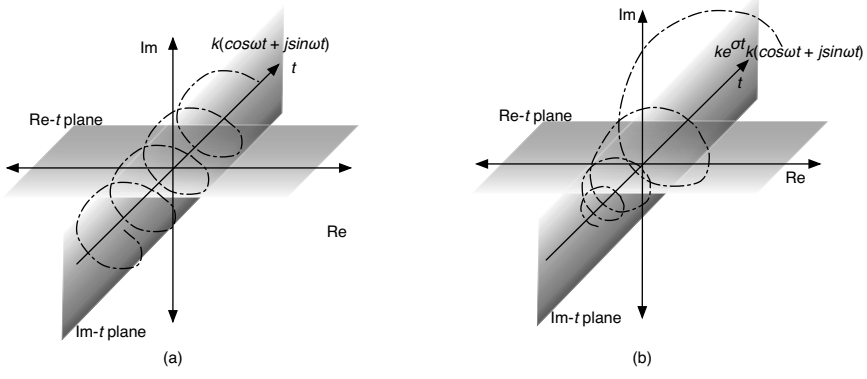


Figure 5.8 More general cases of the complex exponential signal

5.3 Systems

4:55

A system is an entity that acts on a set of input signals to produce a set of output signals. Mathematically speaking, a system is a function whose domain and ranges are both sets of functions. We typically denote the input to the system by the vector function $\boldsymbol{x}(t)$ and its output by the vector function $\boldsymbol{y}(t)$. The **transfer function** $\boldsymbol{H}(.)$ converts the input to the output, so that $\boldsymbol{y}(t) = \boldsymbol{H}(\boldsymbol{x}(t))$.[4]

When studying a system, we have one of two goals in mind: (1) to determine the output from an existing system when it receives a specific input (the **analysis** problem) or (2) to design a system that exhibits a desired transformation from a set of input signals to a set of output signals (the **design** problem). In both cases, we use transforms as an essential analytical and design tool. This chapter focuses on the tools of analysis; Chapter 8, on control theory, focuses on design.

As with signals, systems can be categorized in several ways: **continuous time** and **discrete time**, and **analog** and **digital**. Their definitions mirror those for signals. We discuss four categories in more detail.

Causal systems act only on inputs from the past. That is, they cannot know the future. In contrast, **acausal** systems are aware of the future. Although acausal systems sound improbable, they are rather easy to build if they operate on logs or traces of inputs rather than on inputs presented to the system in real time.

10:38

Memoryless systems act only on the inputs presented at any point in time. By contrast, the output of a **dynamic** system may depend on both current and past inputs.

A **time-invariant** system is one whose parameters do not change with time. So, if the system has an output $\boldsymbol{y}(t)$ at time t for an input $\boldsymbol{x}(t)$, it has the output $\boldsymbol{y}(t - T)$ for an input $\boldsymbol{x}(t - T)$. That is, if $\boldsymbol{y}(t) = \boldsymbol{H}(\boldsymbol{x}(t))$, then $\boldsymbol{y}(t - T) = \boldsymbol{H}(\boldsymbol{x}(t - T))$.

Finally, and most important, a **linear** system is one that exhibits the properties of **additivity** and **homogeneity**.

- A system is **additive** if, given that input $\boldsymbol{x}_1$ leads to output $\boldsymbol{y}_1$ and input $\boldsymbol{x}_2$ leads to output $\boldsymbol{y}_2$, the input $(\boldsymbol{x}_1 + \boldsymbol{x}_2)$ leads to the output $(\boldsymbol{y}_1 + \boldsymbol{y}_2)$. That is, if $\boldsymbol{y}_1 = \boldsymbol{H}(\boldsymbol{x}_1)$ and $\boldsymbol{y}_2 = \boldsymbol{H}(\boldsymbol{x}_2)$, then $\boldsymbol{y}_1 + \boldsymbol{y}_2 = \boldsymbol{H}(\boldsymbol{x}_1 + \boldsymbol{x}_2)$.
- A system is **homogeneous (scalar)** if, given that the input $\boldsymbol{x}$ leads to the output $\boldsymbol{y}$, the input $k\boldsymbol{x}$, where k is a scalar, leads to the output $k\boldsymbol{y}$. That is, if $\boldsymbol{y} = \boldsymbol{H}(\boldsymbol{x})$, then $k\boldsymbol{y} = \boldsymbol{H}(k\boldsymbol{x})$.

These two conditions can be combined into the single condition of **superposition**. A system exhibits superposition if, given that the input $\boldsymbol{x}_1$ leads to output $\boldsymbol{y}_1$

4. We are using the term *transfer function* loosely. As discussed in Section 8.2.2, in the context of control theory, the transfer function is more precisely described as the Laplace transform of the impulse response of the system. At this stage, however, this loose but intuitively appealing description of the transfer function suffices.

and input $\boldsymbol{x}_2$ leads to output $\boldsymbol{y}_2$, then for all constants k_1 and k_2, the input $(k_1\boldsymbol{x}_1 + k_2\boldsymbol{x}_2)$ leads to the output $(k_1\boldsymbol{y}_1 + k_2\boldsymbol{y}_2)$. That is, if $\boldsymbol{y}_1 = \boldsymbol{H}(\boldsymbol{x}_1)$ and $\boldsymbol{y}_2 = \boldsymbol{H}(\boldsymbol{x}_2)$, then $k_1\boldsymbol{y}_1 + k_2\boldsymbol{y}_2 = \boldsymbol{H}(k_1\boldsymbol{x}_1 + k_2\boldsymbol{x}_2)$.

EXAMPLE 5.6: LINEAR SYSTEM

Is the system defined by the transfer function $H(x) = x^2$ linear?

Solution:

$H(k_1x_1 + k_2x_2) = (k_1x_1 + k_2x_2)^2$, which is not the same as $k_1y_1 + k_2y_2 = k_1^2x_1^2 + k_2^2x_2^2$. Hence, the system is not linear.

Note that any system that limits the output to some maximum or minimum is not linear, because such a system is not homogeneous. But in practice, outputs are almost always bounded—that is, they *saturate* at some limiting value—due to physical limitations. Therefore, most real systems are not linear. However, it is often possible to model the system behavior, around its operating point, as being approximately linear. Specifically, we can use the first-order Taylor expansion of the transfer function in the region of the operating point to approximate the behavior of the system. This technique, called **linearization**, is discussed in Section 8.2.2.

5.4 Analysis of a Linear Time-Invariant System



11:46

The class of linear time-invariant (**LTI**) systems is important both because it is relatively easy to analyze and because many common systems can be approximated by LTI systems. Here, we study three aspects of an LTI system: the effect of an LTI system on a complex exponential input, the output of an LTI system with a zero input, and the output of an LTI system for an arbitrary input. We will revisit the third topic later to demonstrate the power of using transform domain techniques in analysis.

Note that the discussion here is for continuous time systems. The analysis of discrete time systems is nearly identical, and we defer this discussion to our study of control theory in Chapter 8.

5.4.1 The Effect of an LTI System on a Complex Exponential Input

Consider an LTI system that is described by

$$\boldsymbol{y}(t) = \boldsymbol{H}(\boldsymbol{x}(t))$$

where $\boldsymbol{y}$ and $\boldsymbol{x}$ are, in general, vector functions of time, and $\boldsymbol{H}$ is the transfer function. For simplicity of exposition, in the remainder of this discussion, we will assume that both x and y are scalar functions of time and so will no longer use boldface to represent them. Suppose that the input is a complex exponential function of time, so that $x(t) = ke^{st}$, where s is complex and k is real. Then,

$$y(t) = H(ke^{st}) \quad \textbf{(EQ 5.17)}$$

Now, consider the input $(e^{s\tau})(ke^{st})$, where τ is independent of t. Because the system is LTI and $e^{s\tau}$ is a scalar with respect to t, the corresponding output is $(e^{s\tau})y(t)$. But note that the input $(e^{s\tau})(ke^{st})$ can be rewritten as $ke^{s(t+\tau)}$, which we recognize as a time-shifted input (i.e., t is replaced by $t+\tau$). Therefore, the corresponding output, because of time invariance, must be $y(t+\tau)$. This gives us the relationship

$$y(t+\tau) = e^{s\tau}y(t) \quad \textbf{(EQ 5.18)}$$

This must hold for all t, so it is true for $t = 0$, where

$$y(\tau) = e^{s\tau}y(0) \quad \textbf{(EQ 5.19)}$$

Differentiating both sides with respect to τ, we get

$$\frac{dy(\tau)}{d\tau} = sy(0)e^{s\tau} = sy(\tau) \quad \textbf{(EQ 5.20)}$$

where, for the last step, we invoke Equation 5.19. Using standard techniques for the solution of ordinary differential equations, it is easy to show that the unique solution to this differential equation is given by

$$y(t) = f(s)e^{st} \quad \textbf{(EQ 5.21)}$$

where $f(s)$ is an arbitrary function independent of t. This is an important result! It shows that for *any* LTI system, if the input is a complex exponential, so too is the output. It is impressive that we can characterize all LTI systems so simply: After all, we have absolutely no idea what the system looks like on the inside, yet we are able to state how it will respond to a complex exponential input.

We can gain additional insight into the nature of this phenomenon by comparing Equations 5.21 and 5.17. We see that for the special input $x(t) = ke^{st}$, the effect of the LTI system is to act as a scalar multiplier, with the multiplication factor being $f(s)/k$. Therefore, $x(t) = ke^{st}$ is an **eigenfunction** of an LTI system: that is, an input function such that the effect of the system is simply to scale it. The corresponding **eigenvalue** is $f(s)/k$.

We can summarize this discussion by stating that the input $x(t) = ke^{st}$ is an eigenfunction of any LTI system for any value of s. Therefore, if we can represent an arbitrary input as the sum of such eigenfunctions, the corresponding output will be the

sum of the scaled inputs. From this perspective, the transforms we are about to study can be viewed as a way to represent an arbitrary input as the sum of complex exponentials.

5.4.2 The Output of an LTI System with a Zero Input


11:48

A wide range of LTI systems can be described in the form of a **linear differential equation**, whose general form is

$$\frac{d^N y(t)}{dt^N} + a_1 \frac{d^{N-1} y(t)}{dt^{N-1}} + \ldots + a_{N-1} \frac{dy(t)}{dt} + a_N y(t) = b_{N-M} \frac{d^M x(t)}{dt^M} + b_{N-M+1} \frac{d^{M-1} x(t)}{dt^{M-1}} + \ldots + b_{N-1} \frac{dx(t)}{dt} + b_N x(t) \quad \text{(EQ 5.22)}$$

In nearly all practical systems, $M > N$. Here, we study the output of this class of LTI system when its input $x(t)$ is zero, so that the right-hand side is zero. Using the operator D to denote differentiation, with D^N representing $\frac{d^N}{dt^N}$, we can rewrite the equation, for zero input, as

$$(D^N + a_1 D^{N-1} + \ldots + a_{N-1} D + a_N) y(t) = 0$$

where the polynomial in D on the left-hand side is called the **characteristic polynomial** of the system. Note that this characteristic polynomial is of degree N. Therefore, from the fundamental theorem of algebra, it can be factorized as

$$((D - \lambda_1)(D - \lambda_2) \ldots (D - \lambda_N)) y(t) = 0$$

where the λ_i (in general complex quantities) are the roots of the polynomial, and some of the roots may be repeated. Assume for the moment that the roots are distinct. In this case, each solution of this equation (that is, a value of $y(t)$ for which the equation is true) is given by

$$(D - \lambda_i) y(t) = 0$$

which we can expand as

$$\frac{dy(t)}{dt} = \lambda_i y(t)$$

Comparing this with Equation 5.20, we see that the solution of this equation is given by

$$y(t) = c_i e^{\lambda_i t}$$

where c is a constant independent of t but may be a function of λ_i. Because each root of the equation generates a solution of this form, and the system is linear, by superposition the general solution (assuming distinct roots) is given by their sum:

$$y(t) = \sum_{i=1}^{N} c_i e^{\lambda_i t} \qquad \textbf{(EQ 5.23)}$$

We interpret this as follows: If a linear differential LTI system, irrespective of its internal organization, receives no input and if the roots of its characteristic polynomial are distinct, its output can be expressed as the sum of complex exponentials. Recall that any complex exponential has an intrinsic frequency. Each root of Equation 5.23 corresponds to a **natural frequency** of the LTI system: a frequency at which the system vibrates given zero input. The solution in Equation 5.23 is therefore also called the **natural response** of the system. If a system is subjected to an input that then ceases, such as a burst of packets entering a router's buffer, the subsequent response of the system, because there is no additional input, will be its natural response. The natural response is a combination of its natural frequencies, which are intrinsic to the system.

A system that is given an input at a frequency that coincides with one of its natural frequencies will *resonate*, amplifying the input. This important phenomenon is widely employed in practical systems. For example, the basis of radio reception is to tune the resonant frequency[5] of an electronic circuit so that it selectively amplifies signals corresponding to a desired transmitter, ignoring the rest of the radio spectrum.

EXAMPLE 5.7: REAL NATURAL RESPONSE OF AN LTI SYSTEM

Compute the natural response of the LTI system given by

$$\frac{d^2y(t)}{dt^2} + \frac{5dy(t)}{dt} + 6y(t) = 8x(t).$$

Solution:

The natural response is given by the differential equation $(D^2 + 5D + 6)y(t) = 0$, which can be factored as $((D+3)(D+2))y(t) = 0$. Thus, the natural response is given by $c_1e^{-3t} + c_2e^{-2t}$, where the two constants can be determined from the initial conditions $y(0)$ and $\dot{y}(0)$. The two phasors corresponding to this solution have no complex component, and therefore both correspond to a natural frequency of 0 (i.e., do not have an oscillatory behavior).

---

5. Strictly speaking, we tune the receiver to a range centered on the resonant frequency.

EXAMPLE 5.8: COMPLEX NATURAL RESPONSE OF AN LTI SYSTEM

Compute the natural response of the LTI system given by $\frac{d^2y(t)}{dt^2} + y(t) = 8x(t)$.

Solution:

The natural response is given by the differential equation $(D^2 + 1)y(t) = 0$, which can be factored as $((D + j)(D - j))y(t) = 0$. Thus, the natural response is given by $c_1e^{-jt} + c_2e^{jt}$, where the two constants can be determined from the initial conditions $y(0)$ and $\dot{y}(0)$. To get more insight into this solution, we use Euler's equation to expand this solution as

$$\begin{aligned} & c_1(\cos(-t) + j\sin(-t)) + c_2(\cos(t) + j\sin(t)) \\ &= c_1(\cos(t) - j\sin(t)) + c_2(\cos(t) + j\sin(t)) \\ &= (c_1 + c_2)\cos(t) + j(c_2 - c_1)\sin(t) \end{aligned}$$

For this solution to be real, it is necessary that $(c_1 + c_2)$ have no imaginary component and that $(c_2 - c_1)$ have no real component. This is possible only if $c_1 = a + jb$ and $c_2 = a - jb$, where a and b are real numbers. Recall that pairs of numbers of this form are called **complex conjugate pairs**. In general, for practical systems, the constants associated with the natural response come in complex conjugate pairs.

Continuing with our solution, we therefore substitute $a \pm jb$ for the constants to get the natural response as $2a\cos(t) + 2b\sin(t)$. We recognize this as an *oscillatory* response, with an angular frequency of 1 radian/second. Interestingly, this natural response does not decay: The system oscillates indefinitely at this frequency of its own accord!

We have thus far assumed that the roots of the characteristic equation are distinct. This need not necessarily be the case. For example, if the characteristic equation is $D^2 + 2D + 1$, the roots are –1 and –1, which are repeated. In the case of repeated roots, the form of the solution is slightly different. For a root λ that is repeated r times, it can be shown that the corresponding solution is given by

$$y(t) = (c_1e^{\lambda t} + c_2te^{\lambda t} + \ldots + c_rt^{r-1}e^{\lambda t}) \quad \textbf{(EQ 5.24)}$$

5.4.3 The Output of an LTI System for an Arbitrary Input



9:28

We now study how an LTI system responds to a nonzero input. We use the notation $x(t) \rightarrow y(t)$ to denote that if the input to a system is $x(t)$, its output is $y(t)$. Suppose

that we are given $h(t)$, the **impulse response** of the system. This is the response to the system when the input is the Dirac delta $\delta(t)$. Using our notation, we write

$$\delta(t) \rightarrow h(t) \qquad \textbf{(EQ 5.25)}$$

What does the impulse response look like? By definition, because it is a delta function, the input ceases immediately after time 0. Therefore, the response of the system is its natural response (other than at time 0 itself). We have already seen how to compute this response.

Recall from Equation 5.13 that multiplying $\delta(t)$ by $x(t)$ results in scaling the impulse by the value $x(0)$; that is, $x(t)\delta(t) = x(0)\delta(t)$. Because the system is linear, the response of the system to the scaled impulse $x(0)\delta(t)$ will be the scaled output $x(0)h(t)$, so that

$$x(t)\delta(t) \rightarrow x(0)h(t). \qquad \textbf{(EQ 5.26)}$$

Recall from Equation 5.14 that if we time-shift the impulse, we select a different "slice" of $x(t)$. Specifically, $x(t)\delta(t-\tau) = x(\tau)\delta(t-\tau)$. Because the system is both linear and time-invariant, the response to this time-shifted scaled impulse $x(\tau)\delta(t-\tau)$ will be the scaled and time-shifted impulse response $x(\tau)h(t-\tau)$, so that

$$x(\tau)\delta(t-\tau) \rightarrow x(\tau)h(t-\tau). \qquad \textbf{(EQ 5.27)}$$

Finally, Equation 5.15 tells us that we can assemble $x(t)$ by integrating these small "slices" together, so that $x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau$. Clearly, this will result in the integration of the corresponding responses together as follows:

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau \rightarrow \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau = x(t) \otimes h(t) \qquad \textbf{(EQ 5.28)}$$

This important result tells us that if we know the impulse response $h(t)$ of an LTI system, we can compute its response to *any* input $x(t)$ by convolving the input signal with the impulse response. Therefore, it is important to be able to compute the convolution of any two functions. This, however, is a difficult and complicated operation, as we saw in Section 5.1.5. An important outcome of transform domain techniques is to convert the difficult convolution operation to a simple multiplication.

5.4.4 Stability of an LTI System

We briefly consider the stability of an LTI system. This topic is covered in more detail in Section 8.8. Intuitively, a system is stable if, when all inputs are removed,

the output either eventually dies down or, at most, oscillates with a bounded amplitude. Here, we study how to characterize the stability of an LTI system.

We have seen that the behavior of an LTI system is described by its natural frequencies, which are expressed as complex exponentials. In Section 5.2.1, we saw that a complex exponential can show only one of three behaviors over time: grow without bound, decay to zero, or oscillate with a constant amplitude and frequency. Therefore, an LTI system can be characterized as stable, oscillatory (or marginally stable), or unstable.

Consider the complex exponential $ke^{(\sigma+j\omega)t}$, which represents a root: a solution to an LTI system's characteristic polynomial. If $\sigma > 0$, this exponential grows without bound. On the other hand, if $\sigma < 0$, the exponential decays to zero. Finally, if $\sigma = 0$, the exponential reduces to a pure sinusoidal oscillation.

Given this observation, we can now easily characterize the stability of an LTI system. If *all* its roots have a value of $\sigma < 0$, the system is **asymptotically stable**. If even one root has a value of $\sigma > 0$, it is **unstable**. Finally, if all the values of σ are 0, the behavior of the system depends on whether there are repeated roots. If there are no repeated roots, the system is **purely oscillatory**. On the other hand, if there are repeated roots, the system is unstable.

This observation allows us great insight into the design of stable LTI systems: We must ensure that the system is such that all the roots of the characteristic polynomial (also called the **poles** of the system) lie in the left half of the complex plane. This lies at the heart of control theory and we will consider it in more detail in Chapter 8.

5.5 Transforms

We are finally ready to embark on a study of transforms. Before plunging in, it is worth recapitulating why transforms are important. Transforms allow us to achieve three goals.

1. The convolution of two functions, which arises in the computation of the output of an LTI system, can be computed more easily by transforming the two functions, multiplying the transformed functions, and then computing the inverse transform.
2. Transforms convert a linear differential equation into a simpler algebraic equation.
3. They give insight into the natural response of a system: the natural frequencies at which it oscillates. This allows us to quickly determine whether there are special frequencies at which the system becomes unstable, whether there are special frequencies that cause the system output to become zero, and the frequencies at which most of the output magnitude can be found.

Table 5.1 Summary of Transforms Described in This Chapter

Transform	Signal Type	Transform Description
Fourier series	Continuous, periodic	Sum of sinusoids
Fourier	Continuous, periodic or aperiodic	Integral of sinusoids
Discrete-time Fourier	Discrete, periodic or aperiodic	Sum of sinusoids
Laplace	Continuous, periodic or aperiodic	Integral of complex exponentials
Z	Discrete, periodic or aperiodic	Sum of complex exponentials

We will consider five transforms: the Fourier series, the Fourier transform, the discrete Fourier transform, the Laplace transform, and the Z transform. All the transforms share some common properties. For now, we observe that a transform converts a function of time into either an infinite sum or an integral of exponentials of a complex quantity. The transforms differ in whether the complex quantity is a pure sinusoid or an exponentially modulated sinusoid and whether the input signal is continuous or discrete, as shown in Table 5.1. Although this may appear confusing at first glance, these concepts should become clear by the end of this chapter.

5.6 The Fourier Series

8:29

We first study **eternal periodic** signals, that is, signals that last for an infinitely long duration and whose values repeat identically after every period of duration T_0. All signals are, of course, time-limited, so this signal cannot be achieved in practice. However, they are still worth studying because the insight gained from the Fourier series allows us to define the Fourier transform for aperiodic signals.

Fourier showed that nearly every periodic signal $x(t)$, other than some pathological signals, can be represented at nearly every value of t as an infinite sum of sinusoids, as follows:

$$x(t) = a_0 + \sum_{k=1}^{\infty} (a_k \cos k\omega_0 t + b_k \sin k\omega_0 t) \qquad \textbf{(EQ 5.29)}$$

where the **fundamental angular frequency** ω_0 is given by

$$\omega_0 = \frac{2\pi}{T_0} \qquad \textbf{(EQ 5.30)}$$

and the constants a_0, a_k, and b_k are real numbers uniquely determined by $x(t)$. This remarkable equation shows that *any* periodic function, not just a function that looks sinusoidal, can be represented as the sum of sinusoids. It is as fundamental as the observation that any natural number can be denoted using only the ten symbols 0–9. More formally, using the appropriate vector space, we can show that the set of pairs of sinusoids in a Fourier series form an infinite orthogonal basis set, so that every vector—corresponding to an eternal periodic function—can be represented as a linear combination of this basis.

Each pair of terms in the series has a frequency that is a multiple of the fundamental frequency and is therefore called a **harmonic** of that frequency. Once the potentially infinite set of values associated with the constants a_0, a_k, and b_k is known, the function itself is completely specified and can be synthesized from them alone. The accuracy of representation quickly improves with the number of terms.[6]

Note that the constant a_0 can be viewed as a degenerate sinusoid with zero frequency. It is called the **DC** component of the signal by analogy to a direct-current electrical system that, unlike an alternating-current, or AC, system, does not have a sinusoidally oscillating voltage or current.

A graphical interpretation of a Fourier series may add some additional insight. Recall from Section 5.1.1 that a sinusoid can be thought of as a being generated by a rotating phasor. We see that each harmonic in the Fourier series corresponds to two rotating phasors that are offset by 90 degrees and with a common rotational frequency of $k\omega_0$ with magnitudes of a_k and b_k, respectively. The sinusoid generated by these phasors add up in just the right way to form $x(t)$.

So far, we have studied only the form of the Fourier series. We do, of course, need a way to determine the constants corresponding to each term in a Fourier series. It can be shown that these are given by the following equations:

$$\begin{aligned} a_0 &= \frac{1}{T_0}\int_0^{T_0} x(t)dt \\ a_k &= \frac{2}{T_0}\int_0^{T_0} x(t)\cos k\omega_0 t dt \\ b_k &= \frac{2}{T_0}\int_0^{T_0} x(t)\sin k\omega_0 t dt \end{aligned} \qquad \textbf{(EQ 5.31)}$$

where the integral is taken over any period of length T_0 (because they are all the same).

6. However, note that due to the **Gibb's phenomenon**, the Fourier series can have an error of about 10%, even with many tens of terms, when representing a sharply changing signal, such as a square wave.

The form of the Fourier series presented so far is called the **sinusoidal form**. These sinusoids can also be expressed as complex exponentials, as we show next. Note that the kth term of the Fourier series is

$$x_k(t) = a_k \cos k\omega_0 t + b_k \sin k\omega_0 t$$


20:19

Using Equations 5.5 and 5.6, we can rewrite this as

$$x_k(t) = \frac{a_k}{2}(e^{jk\omega_0 t} + e^{-jk\omega_0 t}) + \frac{b_k}{2j}(e^{jk\omega_0 t} - e^{-jk\omega_0 t})$$

Collecting like terms, and defining $c_0 = a_0$, $c_k = \frac{1}{2}(a_k - jb_k)$, $c_{-k} = \frac{1}{2}(a_k + jb_k)$, $k>0$, we can rewrite this as

$$x(t) = c_0 + \sum_{k=1}^{\infty} c_k e^{jk\omega_0 t} + \sum_{k=1}^{\infty} c_{-k} e^{j(-k)\omega_0 t}$$

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t} \qquad \textbf{(EQ 5.32)}$$

This compact notation shows that we can express $x(t)$ as an infinite sum of complex exponentials. Specifically, it can be viewed as a sum of an infinite number of phasors, each with a real magnitude c_k and a rotational frequency of $k\omega_0$. It can be shown that the constants c_k can be found, if we are given $x(t)$, by the relation

$$c_k = \frac{1}{T_0}\int_0^{T_0} x(t) e^{-jk\omega_0 t} dt \qquad \textbf{(EQ 5.33)}$$

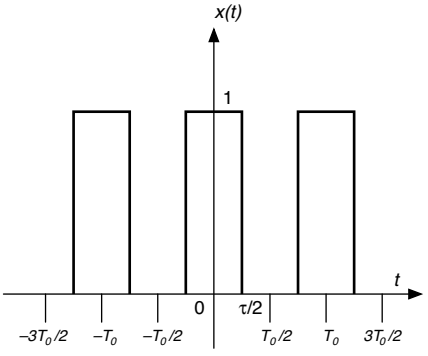
EXAMPLE 5.9: FOURIER SERIES

Find the Fourier series corresponding to the series of rectangular pulses shown in Figure 5.9.

Solution:

The kth coefficient of the Fourier series corresponding to this function is given by

$$c_k = \frac{1}{T_0}\int_0^{T_0} x(t) e^{-jk\omega_0 t} dt\,.$$



1:52

Figure 5.9 An infinite series of rectangular pulses

Instead of choosing the limits from 0 to T_0, we will choose the limits from $-T_0/2$ to $T_0/2$ because it is symmetric about 0. Note that in this range, the function is 1 in the range $[-\tau/2, \tau/2]$ and 0 elsewhere. Therefore, the integral reduces to

$$c_k = \frac{1}{T_0}\int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} e^{-jk\omega_0 t}dt = -\frac{1}{jk\omega_0 T_0}e^{-jk\omega_0 t}\Big|_{-\frac{\tau}{2}}^{\tau/2} = \frac{1}{jk\omega_0 T_0}\left(e^{jk\omega_0\frac{\tau}{2}} - e^{-jk\omega_0\frac{\tau}{2}}\right) \quad \textbf{(EQ 5.34)}$$

Using Equation 5.6 and multiplying the numerator and denominator by $\tau/2$, we can rewrite this as

$$c_k = \frac{\tau}{T_0}\left(\frac{\sin\left(\frac{k\omega_0\tau}{2}\right)}{\frac{k\omega_0\tau}{2}}\right) \quad \textbf{(EQ 5.35)}$$

Note that the coefficients c_k are real functions of τ (not t), which is a parameter of the input signal. For a given input signal, we can treat τ as a constant. Observing that $T_0 = \frac{2\pi}{\omega_0}$, the coefficients can be obtained as the values taken by a continuous function of ω, defined as

$$X(\omega) = \frac{\tau\omega_0}{2\pi}\frac{\sin\left(\frac{\omega\tau}{2}\right)}{\frac{\omega\tau}{2}} \quad \textbf{(EQ 5.36)}$$

for the values $\omega = k\omega_0$. That is, $c_k = X(k\omega_0)$. Thus, c_k is a *discrete* function of ω defined at the discrete frequency values $\omega = k\omega_0$, as shown in Figure 5.10.

The function $sin(x)/x$ arises frequently in the study of transforms and is called the **sinc** function. It is a sine wave whose value at 0 is 1 and elsewhere is the sine function linearly modulated by distance from the *Y*-axis. This function forms the envelope of the Fourier coefficients, as shown in Figure 5.10. Note that the function is zero (i.e., has zero-crossings) when a sine function is zero, that is, when $\omega\tau/2 = m\pi$, $m \neq 0$, or $\omega = 2m\pi/\tau$, $m \neq 0$.

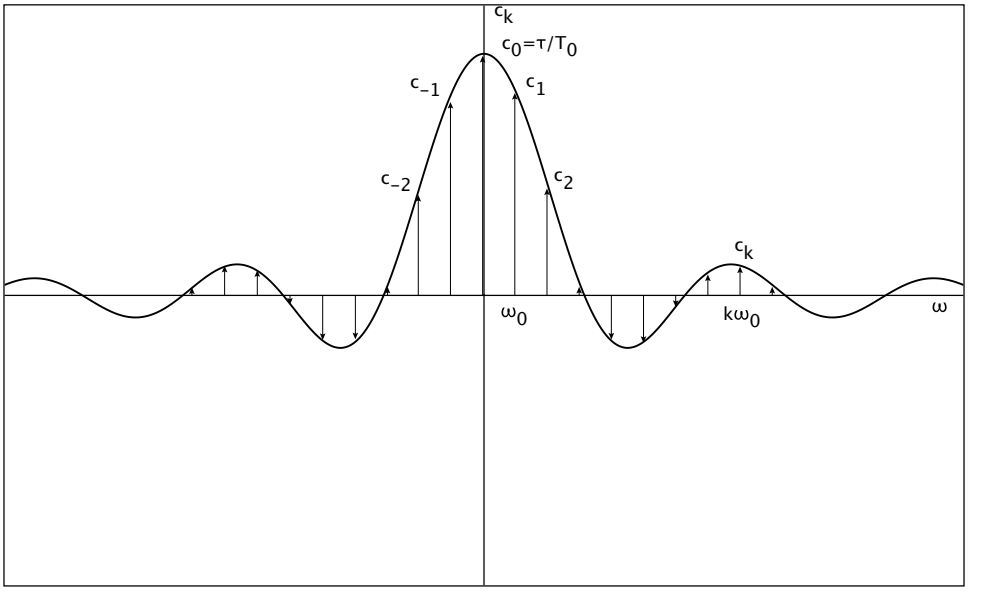


Figure 5.10 The sinc function as a function of ω

5.7 The Fourier Transform and Its Properties


10:55

A Fourier series can represent any *eternal* periodic function as an infinite sum of real sinusoids. However, it cannot be used to represent aperiodic functions. These are represented, instead, using a Fourier transform.

To motivate the Fourier transform, consider the behavior of the pulse series of Example 5.9 when increasing the interpulse spacing T_0 while keeping the pulse width τ constant. Recall that for this signal, the kth Fourier series coefficient c_k is given by

$$X(\omega) = \frac{\tau}{T_0}\frac{\sin\left(\frac{\omega\tau}{2}\right)}{\frac{\omega\tau}{2}} \text{ for } \omega = k\omega_0 \text{ where } \omega_0 = \frac{2\pi}{T_0}.$$

As T_0 increases, ω_0 decreases, which brings the coefficients under the sinc curve, which are spaced ω_0 apart, closer together (see Figure 5.10). In the limit, as $T_0 \to \infty$, the periodic pulse train becomes a single pulse, and the coefficients are spaced infinitely closely together, converging with the sinc curve. Thus, in the limit, the sinc curve corresponds to the Fourier series representing a single pulse and is called its **Fourier transform**.

Arguing along the same lines, we can analyse the limiting behavior of the generic kth Fourier series coefficient

$$c_k = \frac{\omega_0}{2\pi}\int_0^{\frac{2\pi}{\omega_0}} x(t)e^{-jk\omega_0 t}dt$$

as $\omega_0 \to 0$ to define the Fourier transform of a signal $x(t)$ as

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt \qquad \textbf{(EQ 5.37)}$$

This transforms a signal (a function of time) to a function of the complex quantity $j\omega$. The value of the transform for a specific value of $\omega = w$ can be viewed as the amount of signal energy at the frequency w. The magnitude at $\omega = 0$, for example, is the DC component of the signal. We represent the fact that $X(j\omega)$ is the transform of the signal $x(t)$ as

$$x(t) \overset{\mathcal{F}}{\leftrightarrow} X(j\omega) \qquad \textbf{(EQ 5.38)}$$

The double-headed arrow indicates that $x(t)$ can be recovered from its transform, $X(jw)$, using the *inverse Fourier transform* discussed later.

EXAMPLE 5.10: FOURIER TRANSFORM

Compute the Fourier transform of a single rectangular pulse of height 1 and width τ centered on the origin (Figure 5.11). This signal is also denoted $rect(t/\tau)$.

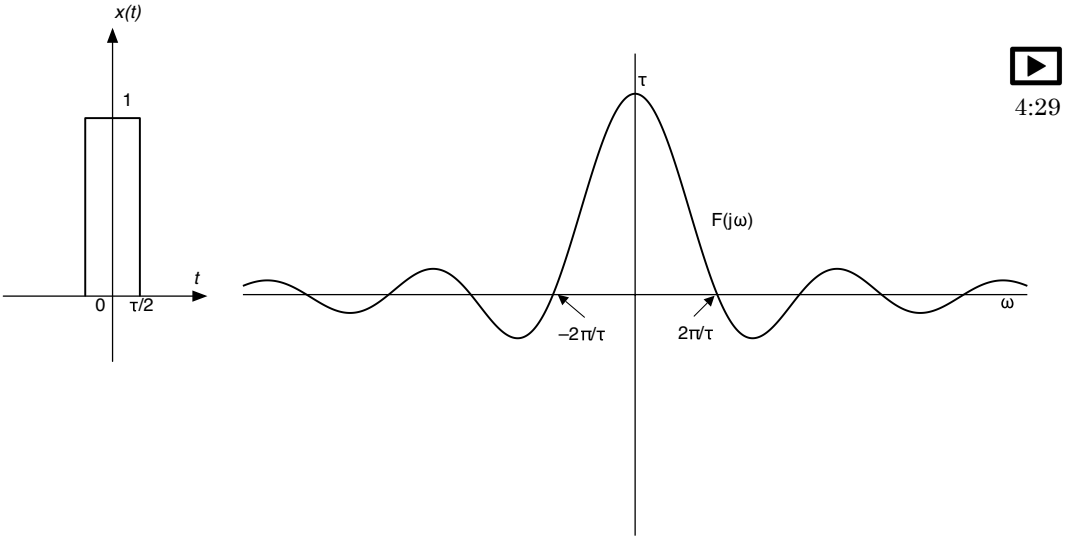


Figure 5.11 A single pulse and its Fourier transform

Solution:

The Fourier transform of this signal $x(t)$ is given by

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt\,.$$

The function is zero outside the interval $[-\tau/2, \tau/2]$ and 1 inside this interval. Therefore, the function reduces to

$$X(j\omega) = \int_{\frac{-\tau}{2}}^{\frac{\tau}{2}} e^{-j\omega t}dt = \frac{-e^{-j\omega t}}{jw}\Big|_{\frac{-\tau}{2}}^{\tau/2} = \frac{e^{j\omega\frac{\tau}{2}} - e^{-j\omega\frac{\tau}{2}}}{j\omega}$$

Multiplying the numerator and denominator by $\frac{\tau}{2}$ as before, and using Euler's formula, we can rewrite this as

$$X(j\omega) = \tau\frac{\sin\left(\frac{\omega\tau}{2}\right)}{\frac{\omega\tau}{2}} = \tau\text{sinc}\left(\frac{\omega\tau}{2}\right) \qquad \textbf{(EQ 5.39)}$$

Therefore, the Fourier transform of $rect\left(\frac{t}{\tau}\right)$ is a sinc function that has a peak of height τ at the origin and zero crossings when $\omega\frac{\tau}{2} = k\pi$, $k \neq 0$: that is, when $\omega = \frac{2k\pi}{\tau}$, as shown in Figure 5.11. Note that the signal has no energy at frequencies corresponding to its zero crossings. This allows us to introduce a narrow-frequency signal at these frequencies and not interfere with the signal. This is the basis of **orthogonal frequency division multiplexing (OFDM)**, which is widely used in such technologies as 4G and WiFi.

The Fourier transform of a signal may not always exist. The existence of a transform is guaranteed by the **Dirichlet conditions**.



7:06

- The signal $x(t)$ is **absolutely integrable**, that is, $\int_{-\infty}^{\infty} |x(t)|dt < \infty$.
- The signal $x(t)$ has a finite number of maxima and minima and a finite number of discontinuities in any finite interval.

These conditions are satisfied by most signals that arise in practical situations. Note that signals with "jumps," such as rectangular pulses, do have Fourier transforms, as we have already seen. Indeed, we can even compute the Fourier transform of a highly discontinuous function, such as an impulse, as in the next example.

EXAMPLE 5.11: FOURIER TRANSFORM OF AN IMPULSE

Compute the Fourier transform of an impulse.

Solution:

The transform is given by

$$X(j\omega) = \int_{-\infty}^{\infty} \delta(t)e^{-j\omega t}dt\,.$$

From Equation 5.15, setting $\tau = 0$, we see that this integral is simply $X(\tau) = e^{-j\omega\tau}\big|_0 = 1$. Thus, the Fourier transform of an impulse is the constant 1, or $\delta(t) \overset{\mathcal{F}}{\leftrightarrow} 1$.

Given a transformed signal, we can recover it by using the inverse Fourier transform, given by

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega \qquad \textbf{(EQ 5.40)}$$

Compare this with Equation 5.32 to see how the summation in the Fourier series is transformed into its limiting integral.

EXAMPLE 5.12: INVERSE FOURIER TRANSFORM

What is the signal corresponding to the transformed signal $\delta(\omega - \omega_0)$?

Solution:

The signal is given by

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \delta(\omega - \omega_0)e^{j\omega t}d\omega\,.$$

But, from Equation 5.15, the integral reduces to $e^{j\omega_0 t}$. Therefore, we have

$$\frac{e^{j\omega_0 t}}{2\pi} \overset{\mathcal{F}}{\leftrightarrow} \delta(\omega - \omega_0), \text{ or } e^{j\omega_0 t} \overset{\mathcal{F}}{\leftrightarrow} 2\pi\delta(\omega - \omega_0).$$

We now turn to some useful properties of the Fourier transform. Proofs of these properties can be derived from first principles, using the definition of the transform, and can be found in standard textbooks on the subject. These properties, along with the table of Fourier transforms in (Table 5.2) allow us to compute the transforms of most functions that arise in practice without having to compute the integral in Equation 5.37 explicitly.

1. **Linearity:** If $x_1(t) \overset{\mathcal{F}}{\leftrightarrow} X_1(j\omega)$ and $x_2(t) \overset{\mathcal{F}}{\leftrightarrow} X_2(j\omega)$, then for any constants a and b, $ax_1(t) + bx_2(t) \overset{\mathcal{F}}{\leftrightarrow} aX_1(j\omega) + bX_2(j\omega)$. Note that a and b can be complex numbers.


3:36

EXAMPLE 5.13: LINEARITY OF THE FOURIER TRANSFORM

Compute the Fourier transform of the eternal sinusoid $\cos(\omega_0 t)$.

Solution:

We use Equation 5.5 to rewrite $\cos(\omega_0 t) = \frac{1}{2}(e^{j\omega_0 t} + e^{-j\omega_0 t})$. From the previous example and the linearity property, we get $\cos(\omega_0 t) \overset{\mathcal{F}}{\leftrightarrow} \pi(\delta(\omega + \omega_0) + \delta(\omega - \omega_0))$. Thus, a sinusoid in the time domain corresponds to two impulses in the frequency domain. These are at the two frequencies ω_0 and $-\omega_0$ and represent counterrotating phasors with the same frequency. The imaginary components of these phasors are always equal and opposite, and their sum is therefore a real sinusoid.

2. **Time-shifting:** If $x(t) \overset{\mathcal{F}}{\leftrightarrow} X(j\omega)$, then $x_1(t - t_0) \overset{\mathcal{F}}{\leftrightarrow} e^{-j\omega t_0} X_1(j\omega)$. That is, shifting the signal forward in time by t_0 units results in multiplying the transform by $e^{-j\omega t_0}$.


3:01

EXAMPLE 5.14: FOURIER TRANSFORM OF A TIME-SHIFTED IMPULSE

Compute the Fourier transform of an impulse that occurs at time t_0.

Solution:

Since $\delta(t) \overset{\mathcal{F}}{\leftrightarrow} 1$, we immediately get $\delta(t - t_0) \overset{\mathcal{F}}{\leftrightarrow} e^{-j\omega t_0}$.

3. **Time-scaling:** If $x(t) \overset{\mathcal{F}}{\leftrightarrow} X(j\omega)$ then $x(at) \overset{\mathcal{F}}{\leftrightarrow} \frac{1}{|a|} X\left(\frac{j\omega}{a}\right)$, where a is an arbitrary constant. Intuitively, $x(at)$ is a times "faster" than $x(t)$. This results in the transform's having a magnitude that is a times smaller and a frequency scale that is a times "slower." For instance, if $a = 5$, what happens at time 10 in the unscaled function happens at time 2 in the scaled function. In contrast, a feature present at 100 radians/second in the unscaled transform is present at 500 radians/second in the scaled transform.

EXAMPLE 5.15: FOURIER TRANSFORM OF A SCALED FUNCTION

Compute the Fourier transform of a rectangular pulse as in Figure 5.11 but with a pulse width of 2τ.

Solution:

The transform of a pulse of width τ is $\tau\text{sinc}\left(\frac{\omega\tau}{2}\right)$. When the pulse is twice as wide, the pulse starts at time $-\tau$ instead of at time $-\frac{\tau}{2}$, expanding the time scale by a factor of 2, so that $a = 1/2$. The transform of the longer pulse is $2\tau\text{sinc}(\omega\tau)$. This has zero crossings when $\omega\tau = k\pi$, that is, when $\omega = \frac{k\pi}{\tau}$, which is twice as fast as before.

---

4. **Duality:** If $x(t) \overset{\mathcal{F}}{\leftrightarrow} X(j\omega)$, then $X(t) \overset{\mathcal{F}}{\leftrightarrow} 2\pi x(-j\omega)$ and $X(jt) \overset{\mathcal{F}}{\leftrightarrow} 2\pi x(-\omega)$. This property allows us to use knowledge of a transform to compute new transforms.

EXAMPLE 5.16: DUALITY

What is the Fourier transform of the constant function $1(t)$?

6:05

Solution:

Since $\delta(t) \overset{\mathcal{F}}{\leftrightarrow} 1$, $1(t) \overset{\mathcal{F}}{\leftrightarrow} 2\pi\delta(-j\omega)$. But $\delta(.)$ is symmetric about the origin, so, we have $1(t) \overset{\mathcal{F}}{\leftrightarrow} 2\pi\delta(j\omega)$, which is an impulse centered on the origin, or at zero frequency. This means that all the energy of the constant function is at a frequency of 0, which is what one would expect.

---

5. **Differentiation:** If $x(t) \overset{\mathcal{F}}{\leftrightarrow} X(j\omega)$, then $\frac{dx^n(t)}{dt^n} \overset{\mathcal{F}}{\leftrightarrow} (j\omega)^n X(j\omega)$.

EXAMPLE 5.17: FOURIER TRANSFORM OF A DERIVATIVE

Compute the Fourier transform of the signal $j\omega_0 e^{j\omega_0 t}$.

Solution:

Since $j\omega_0 e^{j\omega_0 t} = \frac{de^{j\omega_0 t}}{dt}$, then $j\omega_0 e^{j\omega_0 t} \overset{\mathcal{F}}{\leftrightarrow} 2\pi(j\omega)\delta(\omega - \omega_0)$.

Table 5.2 presents some standard transform pairs. These are derived using both first principles and the properties of the Fourier transform discussed earlier.

Table 5.2 Some Standard Fourier Transforms

No.	$x(t)$	$X(j\omega)$	Notes
1	$\delta(t)$	1	
2	1	$2\pi\delta(j\omega)$	
3	$e^{j\omega_0 t}$	$2\pi\delta(\omega-\omega_0)$	
4	$\cos\omega_0 t$	$\pi(\delta(\omega+\omega_0)+\delta(\omega-\omega_0))$	The Fourier transform of a cosine results in impulses in both the positive and negative frequency axes.
5	$\sin\omega_0 t$	$-j\pi(\delta(\omega-\omega_0)-\delta(\omega+\omega_0))$	See preceding note.
6	$u(t)$	$\pi\delta(\omega)+\frac{1}{j\omega}$	
7	$e^{-at}u(t)$	$\frac{1}{a+j\omega}$	$a>0$, and this is an exponential defined only for positive time values.
8	$\cos(\omega_0 t)u(t)$	$\frac{\pi}{2}(\delta(\omega+\omega_0)+\delta(\omega-\omega_0))+\frac{j\omega}{\omega_0^2-\omega^2}$	This is a sinusoid defined only for positive time values.
9	$\sin(\omega_0 t)u(t)$	$\frac{\pi}{2j}(\delta(\omega-\omega_0)-\delta(\omega+\omega_0))+\frac{\omega_0}{\omega_0^2-\omega^2}$	This is a sinusoid defined only for positive time values.
10	$rect\left(\frac{t}{\tau}\right)$	$\tau\text{sinc}\left(\frac{\omega\tau}{2}\right)$	This is a pulse symmetric about the origin with width τ.

See the video link next to 'Duality'

6. **Convolution property:** If

$x_1(t) \overset{\mathcal{F}}{\leftrightarrow} X_1(j\omega)$, $x_2(t) \overset{\mathcal{F}}{\leftrightarrow} X_2(j\omega)$, $y(t) \overset{\mathcal{F}}{\leftrightarrow} Y(j\omega)$, and $y(t) = x_1(t) \otimes x_2(t)$, then $Y(jw) = X_1(jw)X_2(jw)$. That is, convolution in the time domain corresponds to multiplication in the transform domain. For an LTI system with an impulse response of $h(t)$, we know from Equation 5.28 that an input of $x(t)$ leads to an output of $y(t) = x(t) \otimes h(t)$. Therefore, $Y(jw) = X(jw)H(jw)$. We can recover $y(t)$ from this by taking the inverse transform.

The symmetric convolution property also holds: Multiplication in the time domain corresponds to convolution in the transform domain. That is, if

$x_1(t) \overset{\mathcal{F}}{\leftrightarrow} X_1(j\omega)$, $x_2(t) \overset{\mathcal{F}}{\leftrightarrow} X_2(j\omega)$, $y(t) \overset{\mathcal{F}}{\leftrightarrow} Y(j\omega)$, and $y(t) = x_1(t)x_2(t)$,

then $Y(jw) = \frac{1}{2\pi} X_1(jw) \otimes X_2(jw)$.

EXAMPLE 5.18: USING CONVOLUTION TO SOLVE A SYSTEM

Consider a system such that the Fourier transform $H(j\omega)$ of its transfer function $h(t)$ is $\frac{1}{j\omega + 2}$. What is its response to the input $e^{-t}u(t)$?

Solution:

Because $x(t) = e^{-t}u(t)$, from row 7 of Table 5.2, $X(j\omega) = \frac{1}{1 + j\omega}$. Therefore, $Y(j\omega) = \frac{1}{(1 + j\omega)(2 + j\omega)}$. Using the method of partial fractions (i.e., writing the expression as $\frac{a}{1 + j\omega} + \frac{b}{2 + j\omega}$ and solving for a and b),[7] we find that $Y(j\omega) = \frac{1}{1 + j\omega} - \frac{1}{2 + j\omega}$. To find the inverse transform, recall that the Fourier transform is linear, so we need to find only the inverse transform of each term in isolation. From row 7 of Table 5.2 again, we get $y(t) = (e^{-t} - e^{-2t})u(t)$.

---

7. More generally, for $F(x) = \frac{P(x)}{(x - \lambda_1)(x - \lambda_2)\ldots(x - \lambda_r)}$, we can write

$F(x) = \frac{k_1}{(x - \lambda_1)} + \ldots + \frac{k_r}{(x - \lambda_r)}$, where $k_r = [(x - \lambda_r)F(x)]\big|_{x = \lambda_r}$. Also see Section 8.11.

It is clear that the Fourier transform greatly simplifies the analysis of LTI systems: Instead of having to deal with the complex convolution operator, we need merely to deal with multiplying two transforms and then taking the inverse. In general, we can always find the inverse of the product from first principles by using Equation 5.40. However, because the Fourier transform may not always exist, system analysis is typically performed using the Laplace transform.

This concludes our discussion of the Fourier transform. We now turn our attention to the more general Laplace transform, which shares most of the properties of the Fourier transform and can also be applied to a wider range of functions.

5.8 The Laplace Transform



The Fourier transform gives us insight into the frequencies in which a signal's energy is concentrated and simplifies the computation of convolutions. Its main problem is that it is not guaranteed for functions that do not satisfy the Dirichlet conditions. We now study the Laplace transform, which shares most of the properties of the Fourier transform and, additionally, is defined for a wider class of functions.

Recall that the Fourier transform is defined by $X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$. The Laplace transform is defined by

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt \qquad \textbf{(EQ 5.41)}$$

where $s = \sigma + j\omega$. Intuitively, instead of taking the limiting sum of a series of pure sinusoids, as with the Fourier transform, the Laplace transform takes the limiting sum of a series of complex exponentials. Alternatively, writing e^{-st} as $e^{-\sigma t}e^{-j\omega t}$, we can write

$$X(s) = \int_{-\infty}^{\infty} (x(t)e^{-\sigma t})e^{-j\omega t}dt \qquad \textbf{(EQ 5.42)}$$

so that the Laplace transform is identical to the Fourier transform of the function $(x(t)e^{-\sigma t})$. Intuitively, given a signal $x(t)$ that does not satisfy the Dirichlet conditions, we multiply it by an exponentially decreasing function $e^{-\sigma t}$, so that the product becomes absolutely integral. Note that, in this process, we must be careful to select a value of σ large enough to ensure convergence. Therefore, unlike the Fourier transform, in specifying the Laplace transform, it is essential to give the **region of convergence**, that is, the values of σ for which the Laplace transform is defined.

As with the Fourier transform, we denote the relationship between a signal $x(t)$ and its transform $X(s)$ by using the notation

$$x(t) \overset{\mathcal{L}}{\leftrightarrow} X(s) \qquad \textbf{(EQ 5.43)}$$

Note that the Laplace transform of a signal is, in general, a *complex* function, with both real and imaginary parts, hence its representation as $X(s)$.

The inverse Laplace transform is given by

$$x(t) = \frac{1}{2\pi j} \int_{(\sigma - j\infty)}^{(\sigma + j\infty)} X(s) e^{st} ds \qquad \textbf{(EQ 5.44)}$$

The limits of the integral denote integrating from $-\infty$ to ∞ along the imaginary axis for a value of σ chosen in the transform's region of convergence. In practice, the inverse operation is rarely performed from first principles, because the inverse can be found from a table of standard transforms and the (nice!) properties of the Laplace transform.

5.8.1 Poles, Zeroes, and the Region of Convergence


11:30

We now examine in greater depth the somewhat mysterious notion of a region of convergence by referring to a concrete example. Observe that the unit step signal $u(t)$ is not absolutely integrable because the area under this signal is infinite. Therefore, it does not satisfy the Dirichlet conditions, and a Fourier transform is not guaranteed. Nevertheless, it can be shown that its Fourier transform is $\pi\delta(\omega) + \frac{1}{j\omega}$. In the next example, we will compute its Laplace transform.

EXAMPLE 5.19: LAPLACE TRANSFORM OF A UNIT STEP

Compute the Laplace transform of the unit step signal.

Solution:

By definition,

$$X(s) = \int_{-\infty}^{\infty} u(t) e^{-st} dt = \int_{0}^{\infty} e^{-st} dt = \left. \frac{e^{-st}}{-s} \right|_{0}^{\infty}.$$

Now, $e^{-s\infty} = 0$ if $Re(s) > 0$. Assuming that this is the case, we can evaluate the integral as

$$\frac{1}{s} \qquad Re(s) > 0\,.$$

Thus,

$$u(t) \overset{\mathscr{L}}{\leftrightarrow} \frac{1}{s} \qquad Re(s) > 0\,.$$

The region of convergence of this transform is the set of all values of s where the condition $Re(s) > 0$ holds. Recall that s is complex, so this is the right-half plane of the complex plane.

The next example reinforces this notion.

EXAMPLE 5.20: LAPLACE TRANSFORM OF A REAL EXPONENTIAL

Compute the Laplace transform of the signal $x(t) = u(t)e^{at}$, where a is a real constant.

Solution:

By definition,

$$X(s) = \int_{-\infty}^{\infty} u(t)e^{at}e^{-st}dt = \int_{0}^{\infty} e^{-(s-a)t}dt = \left.\frac{e^{-(s-a)t}}{-(s-a)}\right|_{0}^{\infty}.$$

As before, $e^{-(s-a)\infty}$ is 0 if $Re(s-a) > 0$, that is, $Re(s) > a$. If this condition holds, the integral evaluates to $\frac{1}{(s-a)}$. Therefore,

$$u(t)e^{-at} \overset{\mathscr{L}}{\leftrightarrow} \frac{1}{(s-a)} \qquad Re(s) > a.$$

In this case, the region of convergence is the complex half-plane defined by $Re(s) > a$.

What if $s = a$? In this case, the denominator becomes zero, and the transform's value is infinite. This is called a **pole** of the system. (The pole in the previous example was at 0.)

It is easy to show that the transform pair is valid even for a complex a, where $a = \sigma + j\omega$, as long as $Re(s) > \sigma$.

The values of s for which the transform vanishes are called the **zeroes** of the transform. This is illustrated by the next example.

EXAMPLE 5.21: LAPLACE TRANSFORM OF A SINUSOID

Compute the Laplace transform of the sinusoid $u(t)\cos\omega_1 t$.

Solution:

We use Euler's formula to rewrite the signal as $u(t)\left(\frac{e^{j\omega_1 t}+e^{-j\omega_1 t}}{2}\right)$. By definition,

$$\begin{aligned} X(s) &= \int_{-\infty}^{\infty} u(t)\left(\frac{e^{j\omega_1 t}+e^{-j\omega_1 t}}{2}\right)e^{-st}dt \\ &= \frac{\left(\int_0^{\infty} e^{j\omega_1 t}e^{-st}dt + \int_0^{\infty} e^{-j\omega_1 t}e^{-st}dt\right)}{2} \\ &= \frac{1}{2}\left(\frac{1}{s-j\omega_1}+\frac{1}{s+j\omega_1}\right), \end{aligned}$$

where, in the last step, we used the result from the previous example, and we are assuming that $Re(s) > 0$. This reduces to $\frac{s}{s^2+\omega_1^2}$, with the region of convergence as $Re(s) > 0$. Note that the transform becomes infinite for $s = \pm j\omega_1$—the poles of the transform—and is zero for $s = 0$, which is the zero of the transform.

Note that the zero lies outide of the region of convergence.

It is important to keep track of the region of convergence of the transform. Two time-domain functions that are completely different may have an identical transform and may differ only in their region of convergence. Therefore, the Laplace transform is unique only if the region is also specified.

If the region of convergence of the Laplace transform of a signal includes the imaginary axis, the Fourier transform of the signal is defined and can be obtained by setting $s = j\omega$. Otherwise, the Laplace transform of the signal exists but not its Fourier transform.

5.8.2 Properties of the Laplace Transform

The Fourier transform is a special case of the Laplace transform. Therefore, almost all the properties of the Fourier transform—namely, linearity, time-shifting, time-scaling, differentiation, and the convolution property—also hold for the Laplace transform, though in a slightly different form. We summarize the corresponding properties in Table 5.3.

Note that the differentiation result in the table is for the bilateral Laplace transform. The unilateral transform of dx(t)/dt = sX(s) - x(0). With frequency scaling, the new ROC is α + Re(a) < Re(s) < β + Re (a).

Table 5.3 Properties of the Laplace Transform

Property	Precondition		Postcondition	Notes
Linearity	$x_1(t) \overset{\mathcal{L}}{\leftrightarrow} X_1(s)$	$\alpha_1 < Re(s) < \beta_1$	$ax_1(t) + bx_2(t) \overset{\mathcal{L}}{\leftrightarrow} aX_1(s) + bX_2(s)$	Both *a* and *b* are arbitrary constants and can be complex.
	$x_2(t) \overset{\mathcal{L}}{\leftrightarrow} X_2(s)$	$\alpha_2 < Re(s) < \beta_2$	$max(\alpha_1, \alpha_2) < Re(s) < min(\beta_1, \beta_2)$	
Time scaling	$x(t) \overset{\mathcal{L}}{\leftrightarrow} X(s)$	$\alpha < Re(s) < \beta$	$x(at) \overset{\mathcal{L}}{\leftrightarrow} \frac{1}{\lvert a \rvert} X\left(\frac{s}{a}\right)$ $a\alpha < Re(s) < a\beta$	A compression in the time scale expands the frequency scale.
Frequency scaling	$x(t) \overset{\mathcal{L}}{\leftrightarrow} X(s)$	$\alpha < Re(s) < \beta$	$\frac{1}{\lvert a \rvert} x\left(\frac{t}{a}\right) \overset{\mathcal{L}}{\leftrightarrow} X(as)$ $\frac{\alpha}{a} < Re(s) < \frac{\beta}{a}$	A compression in the frequency scale expands the time scale.
Time-shifting	$x(t) \overset{\mathcal{L}}{\leftrightarrow} X(s)$	$\alpha < Re(s) < \beta$	$x(t - t_0) \overset{\mathcal{L}}{\leftrightarrow} e^{-st_0} X(s)$ $\alpha < Re(s) < \beta$	Delaying by a time t_0 multiplies the transform by e^{-st_0}.
Frequency-shifting	$x(t) \overset{\mathcal{L}}{\leftrightarrow} X(s)$	$\alpha < Re(s) < \beta$	$e^{at} x(t) \overset{\mathcal{L}}{\leftrightarrow} X(s - a)$	Note the change in the region of convergence due to frequency shifting. $\alpha + Re(a) < Re(s) < \beta + Re(a)$
Differentiation	$x(t) \overset{\mathcal{L}}{\leftrightarrow} X(s)$	$\alpha < Re(s) < \beta$	$\frac{d^n x(t)}{dt^n} \overset{\mathcal{L}}{\leftrightarrow} s^n X(s)$ $\alpha < Re(s) < \beta$	Differentiation in the time domain corresponds to multiplication by a factor of *s* in the transform domain.

continues

Note that this differentiation result is for the bilateral Laplace transform. The unilateral transform of dx(t)/dt = sX(s) - x(0).

Table 5.3 Properties of the Laplace Transform (*Continued*)

Property	Precondition	Postcondition	Notes
Integration	$x(t) \overset{\mathcal{L}}{\leftrightarrow} X(s) \qquad \alpha < Re(s) < \beta$	$\int_{-\infty}^{t} x(r)dr \overset{\mathcal{L}}{\leftrightarrow} \frac{X(s)}{s}$ $max(\alpha, 0) < Re(s) < \beta$	Integration in the time domain corresponds to division by a factor of s in the transform domain. Note that the region of convergence also changes.
Convolution in time domain	$x_1(t) \overset{\mathcal{L}}{\leftrightarrow} X_1(s) \qquad \alpha_1 < Re(s) < \beta_1$ $x_2(t) \overset{\mathcal{L}}{\leftrightarrow} X_2(s) \qquad \alpha_2 < Re(s) < \beta_2$ $y(t) = x_1(t) \otimes x_2(t)$	$y(t) \overset{\mathcal{L}}{\leftrightarrow} Y(s)$ $Y(s) = X_1(s)X(s)_2$ $max(\alpha_1, \alpha_2) < Re(s) < min(\beta_1, \beta_2)$	The value $y(t)$ is a convolution of two functions $x_1(t)$ and $x_2(t)$. The product of their transforms $X_1(s)$ and $X_2(s)$ determine $Y(s)$, the transform of $y(t)$, and its region of convergence.
Multiplication in time domain	$x_1(t) \overset{\mathcal{L}}{\leftrightarrow} X_1(s) \qquad \alpha_1 < Re(s) < \beta_1$ $x_2(t) \overset{\mathcal{L}}{\leftrightarrow} X_2(s) \qquad \alpha_2 < Re(s) < \beta_2$ $y(t) = x_1(t)x_2(t)$	$y(t) \overset{\mathcal{L}}{\leftrightarrow} Y(s)$ $Y(s) = \frac{1}{2\pi j} X_1(s) \otimes X_2(s)$ $\alpha_1 + \alpha_2 < Re(s) < \beta_1 + \beta_2$	The value $y(t)$ is a product of two functions $x_1(t)$ and $x_2(t)$. The convolution of their transforms $X_1(s)$ and $X_2(s)$ determines $Y(s)$, the transform of $y(t)$, and its region of convergence.
Final value theorem	$X(s)$	$\lim_{t \to \infty} x(t) = \lim_{s \to 0} sX(s)$	The limiting value of $x(t)$ in the time domain as $t \to \infty$ is given by finding the limit of $sX(s)$ as $s \to 0$ in the transform domain.

Table 5.4 Some Standard Laplace Transforms

No.	Signal	$x(t)$	$X(s)$	Region of Convergence
1	Delta or unit impulse	$\delta(t)$	1	All s
2	Unit step	$u(t)$	$1/s$	$Re(s) > 0$
3	Delayed delta	$\delta(t - t_0)$	$e^{-t_0 s}$	All s
4	Ramp	$tu(t)$	$\frac{1}{s^2}$	$Re(s) > 0$
5	Exponential decay	$e^{\alpha t}u(t)$	$\frac{1}{s - \alpha}$	$Re(s) > \alpha$
6	*N*th power decay	$\frac{t^n}{n!}e^{\alpha t}u(t)$	$\frac{1}{(s - \alpha)^{n+1}}$	$Re(s) > \alpha$
7	Sine	$\sin(\omega t)u(t)$	$\frac{\omega}{s^2 + \omega^2}$	$Re(s) > 0$
8	Cosine	$\cos(\omega t)u(t)$	$\frac{s}{s^2 + \omega^2}$	$Re(s) > 0$
9	Exponentially modulated sine	$e^{\alpha t}\sin(\omega t)u(t)$	$\frac{\omega}{(s - \alpha)^2 + \omega^2}$	$Re(s) > \alpha$
10	Exponentially modulated cosine	$e^{\alpha t}\cos(\omega t)u(t)$	$\frac{s - \alpha}{(s - \alpha)^2 + \omega^2}$	$Re(s) > \alpha$

These properties, along with the table of common transforms (Table 5.4) allow us to derive the transform of many common signals without having to derive them from first principles. Note that these transforms are defined for functions that exist only for $t > 0$, so that the Laplace integral has limits from 0 to ∞. This is also called the **unilateral** Laplace transform. In a practical system, a *causal* signal is 0 for $t < 0$, so the unilateral transform suffices for all practical systems.

EXAMPLE 5.22: USING THE LAPLACE TRANSFORM TO SOLVE A SYSTEM

Consider a system such that the Laplace transform $H(s)$ of its transfer function $h(t)$ is $\frac{1}{s + 2}$. What is its response to the input $e^{-t}u(t)$?

Solution:

Since $x(t) = e^{-t}u(t)$, from row 5 of Table 5.4, $X(s) = \frac{1}{s+1}$. Therefore, $Y(s) = \frac{1}{(1+s)(2+s)}$. Using the method of partial fractions, i.e., writing the expression as $\frac{a}{1+s} + \frac{b}{2+s}$ and solving for a and b—also see Section 8.11—we find that $Y(s) = \frac{1}{1+s} - \frac{1}{2+s}$. To find the inverse transform, recall that the Laplace transform is linear, so we need only find the inverse transform of each term in isolation. From row 5 of Table 5.4 again, we get $y(t) = (e^{-t} - e^{-2t})u(t)$.

This analysis assumes that the system is at rest at time zero. If this is not the case, the actual response would be the sum of the natural response, which is the way the system behaves if there is no external input, and the forced response, which is the way the system behaves assuming that its initial state, is at rest. The details of this analysis are beyond the scope of this text.

This concludes our study of the Laplace transform. We now focus on two transforms that deal with *discrete* signals rather then continuous signals, as we have done so far.

We will not be covering material beyond this point in the text

5.9 The Discrete Fourier Transform and Fast Fourier Transform

Most signals in computing systems take on values only at discrete moments in time. Specifically, the signal is defined only at times nT, where n is an integer, and T, a real, is the inverse of the clock frequency. We denote such a signal as $x[nT]$ in contrast to a continuous signal, which is denoted $x(t)$.

5.9.1 The Impulse Train

We define an impulse train $s_T(t)$ with parameter T to be a signal that consists of an infinite series of delta signals with time-separation T. An impulse train is denoted as

$$s_T = \sum_{n=-\infty}^{\infty} \delta(t - nT) \qquad \textbf{(EQ 5.45)}$$

What is the Fourier transform of an impulse train? By the linearity property of the Fourier transform, the transform of the infinite sum in Equation 5.45 is the